

OPERATION AND MAINTENANCE MANUAL – PART 7A

EXCITATION SYSTEM – EXCITATION CUBICLE ERECTION INSTRUCTION

DOCUMENT NO.: VIN/90/PCS/400018/EX/OMM007A/A

Plant Name : VINH-SON HYDRO POWER STATION
Location : VIETNAM
Plant Type : STATIC EXCITATION SYSTEM
Plant Rating : 40MVA

A	Revised after Commissioning	2011-10-20	HL Cheng	Davor Tomerlin	Davor Tomerlin
-	Initial Issue	2011-04-20	HL Cheng	Davor Tomerlin	Davor Tomerlin
Rev	Description	Date	Prepared	Checked	Approved
					

© ALSTOM 2011. All rights reserved. Information contained in this document is indicative only. No representation or warranty is given or should be relied on that it is complete or correct or will apply to any particular project. This will depend on the technical and commercial circumstances. It is provided without liability and is subject to change without notice. Reproduction, use or disclosure to third parties, without express written authority, is strictly prohibited.

Copyright © ALSTOM 2008. We reserve all rights in this document and in the information contained therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

SOMMAIRE / TABLE OF CONTENTS

Chapters		Remarks	
		Yes	No
Section 7A	Excitation cubicle erection instruction		
	4		
7A.1	General	4	
7A.2	Safety instructions	4	
7A.3	Description of equipment	5	
7A.4	Checking after delivery	5	
7A.5	Handling the package	5	
7A.6	Unpacking	8	
7A.7	Handling the cubicle	9	
7A.7.1	Slings principle with lifting eyes on the bottom	9	
7A.7.2	Slings principle with lifting eyes on the top	10	
7A.8	Installing the cubicle	11	
7A.9	Wiring	12	
7A.9.1	Necessary documents	12	
7A.9.2	Safety measures	12	
7A.9.3	Electrical connections	12	
7A.9.4	Preparation before wiring	12	
7A.9.5	Wiring	13	
7A.10	Cleaning	14	

Section 7A Excitation cubicle erection instruction

7A.1 General

This document contains full details on the assembly and installation of a ALSPA type excitation system, along with relevant safety instructions.

7A.2 Safety instructions

Before you start to install the equipment, read the safety instructions below carefully. This applies to all installation and connection procedures of the excitation system.

For safety reasons, during handling operations, nobody must stay under hanging loads.

Follow the instructions printed on the wooden case on how to handle the complete package.

Do not unpack the cubicle until you are about to install it on the site.

Use the special lifting eyes provided on the cubicle.

The cubicle should only stand on its base. It should not be turned upside down or otherwise shaken during handling operations.

The cubicle should be handled with an appropriate beam and slings (consistent with the dimensions and weight of the package).

Never use the door handles of the cubicle to move or handle the equipment.

Before you begin wiring the cubicle, read the wiring diagram and check with a voltmeter for absence of voltage.

Do not connect any electrical device to live terminals.

Operators should wear personal safety equipment in compliance with relevant local regulations, at least safety shoes and a helmet.

Personnel should be made aware of the danger of the lifting eyes when in the vicinity of the cubicle.

Work on the excitation cubicle should only be carried out by skilled persons.

7A.3 Description of equipment

The excitation system is contained in a compact cubicle. For access to the components, the cubicle has doors on the front side. Removable panels are fitted on the lateral and back sides of the cubicle, for access to the internal busbars. Openings are provided on the top for the ventilation of the power components.

The cubicle is mounted on a metallic base frame, fitted with fixing holes and with four lifting eyes used to handle the cubicle.

Dimensions: See the cubicle layout drawing of the project for details.

7A.4 Checking after delivery

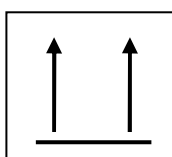
Check the state of the package while it is being unloaded from the truck. If the package shows any damage, make a reservation with the carrier, if possible in presence of an ALSTOM representative. If you have any doubt, do not hesitate to write a reservation record.

7A.5 Handling the package

Handling symbols are printed on the packing case giving instructions about correct handling.

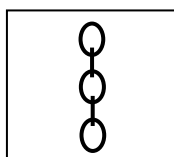
Main indications written on the package

Upright position



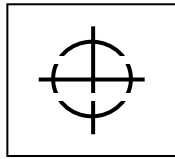
Indicates the correct transport position of the package.

Slings symbol



Shows where slings have to be attached to lift the transport package.

Centre of gravity



Indicates the centre of gravity of the transport package.

Handling the package

The packing case is composed of wooden side panels, a main floor, a second floor, and a hard board for the roof.

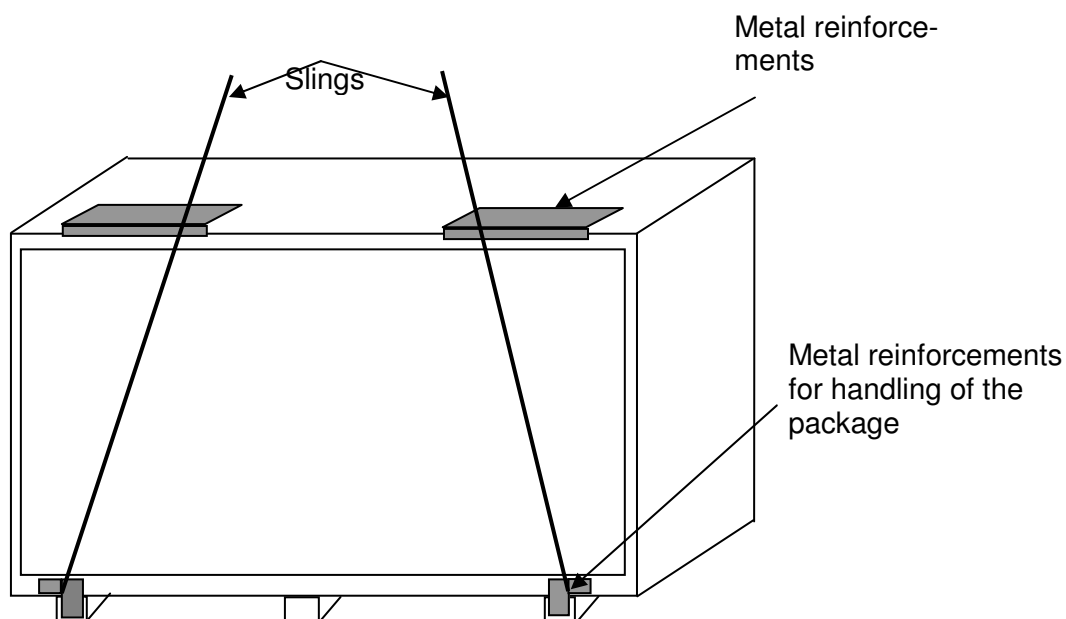
The slinging positions are marked by slinging symbols; the slings should be hooked on the feet, which are fitted with metallic reinforcements.

Keep the package upright while it is handled.

Only skilled persons should handle the package.

The cubicle should not be unpacked unless it has been delivered to the final installation area.

Standard package layout



Storage

Temporary storage on site

To avoid corrosion or other damage, the excitation cubicle should be kept in its original shipping package and stored in a dry, clean and vibration free place. Insects and small animals may cause substantial damage to the equipment, so be careful to keep such animals away from the storage area.

The standard storage time is six months, including the transport time.

The storage temperature range extends from -25°C to +55°C, with max. 85% relative humidity.

Do not expose the package to rain or snow.

Do not clean the cubicle or the package with a water pressure jet.

Conservation on site

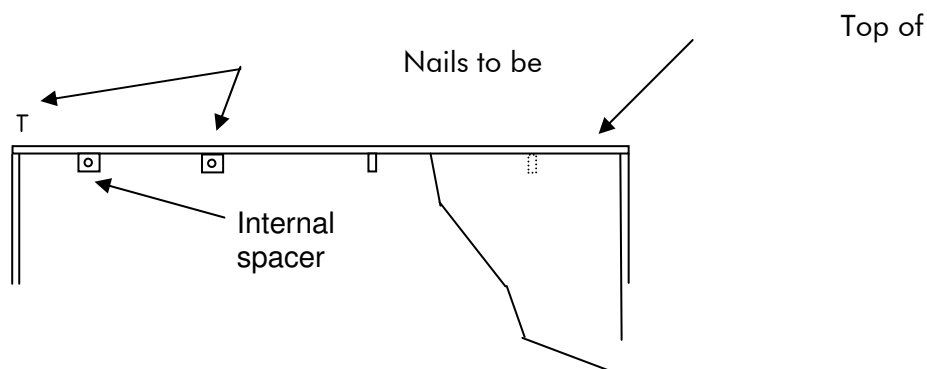
On site, the cubicle should be kept in its protective plastic cover.

The keys for the cubicle doors are sealed in plastic bags attached on the door handles.

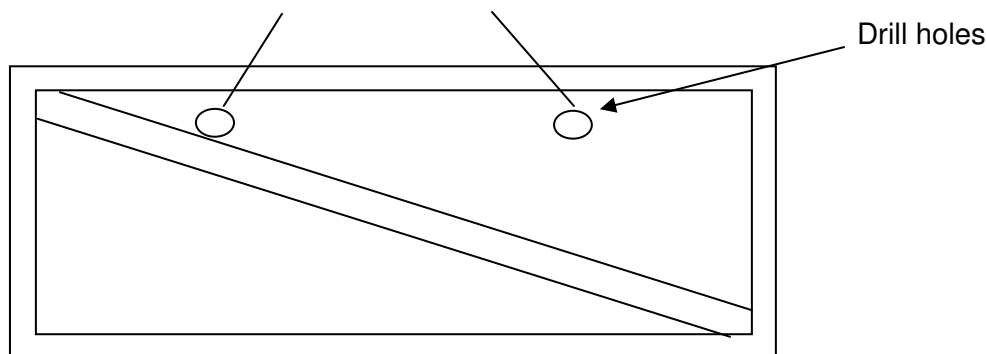
A set of 3 keys is supplied for each door lock.

7A.6 Unpacking

The roof panel of the package is removed first. Extract the nails which clamp the hard board roof to the side panels and to the internal spacer. Remove the roof panel from the package. Take away the internal spacer and the dampers on the top of the cubicle.



Now the side panels of the package can be removed. The package panels may be very heavy (more than hundred kg). Before removing a side panel, it may be necessary to secure it in position to prevent any risk of accidents. For example, you may drill a hole in the panel in order to attach a handling sling. (Be careful not to damage the cubicle while drilling the holes).



Remove all plastic pads glued on the equipment. The cubicle is protected by a plastic cover. This cover should stay in place until the final installation in order to avoid any damage to the cubicle.

Check the cubicle for transport damage. In the event of any damage, make a record as instructed in § 7A.4.

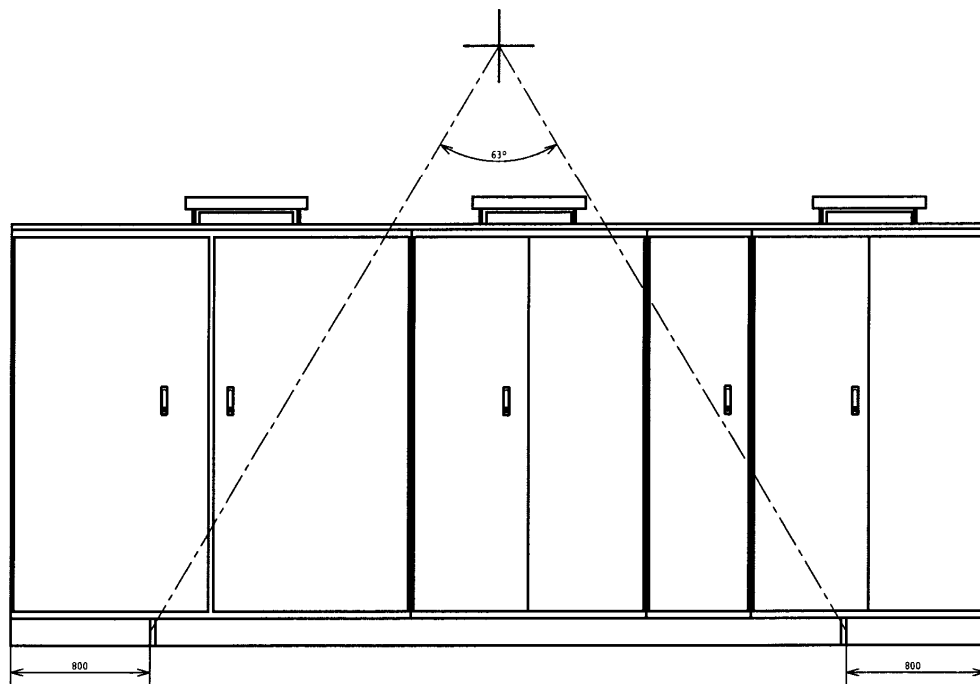
7A.7 Handling the cubicle

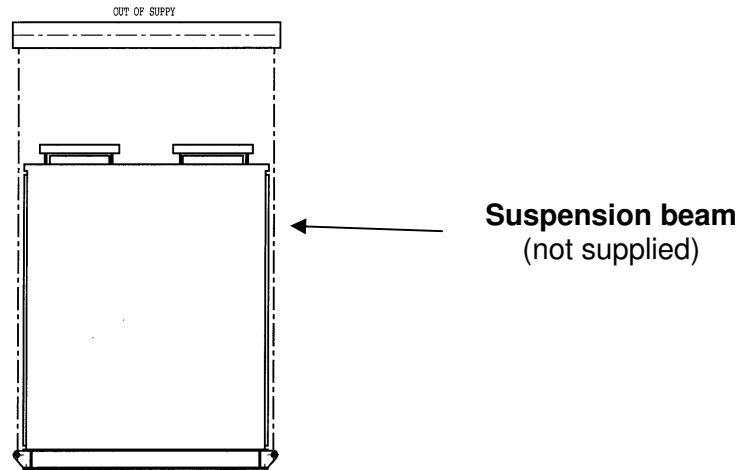
If the cubicle has to be moved, handle it using the lifting eyes on the base frame (refer to the cubicle layout drawing for the location of lifting eyes).

A suspension beam (not supplied) must always be used to lift the cubicle.

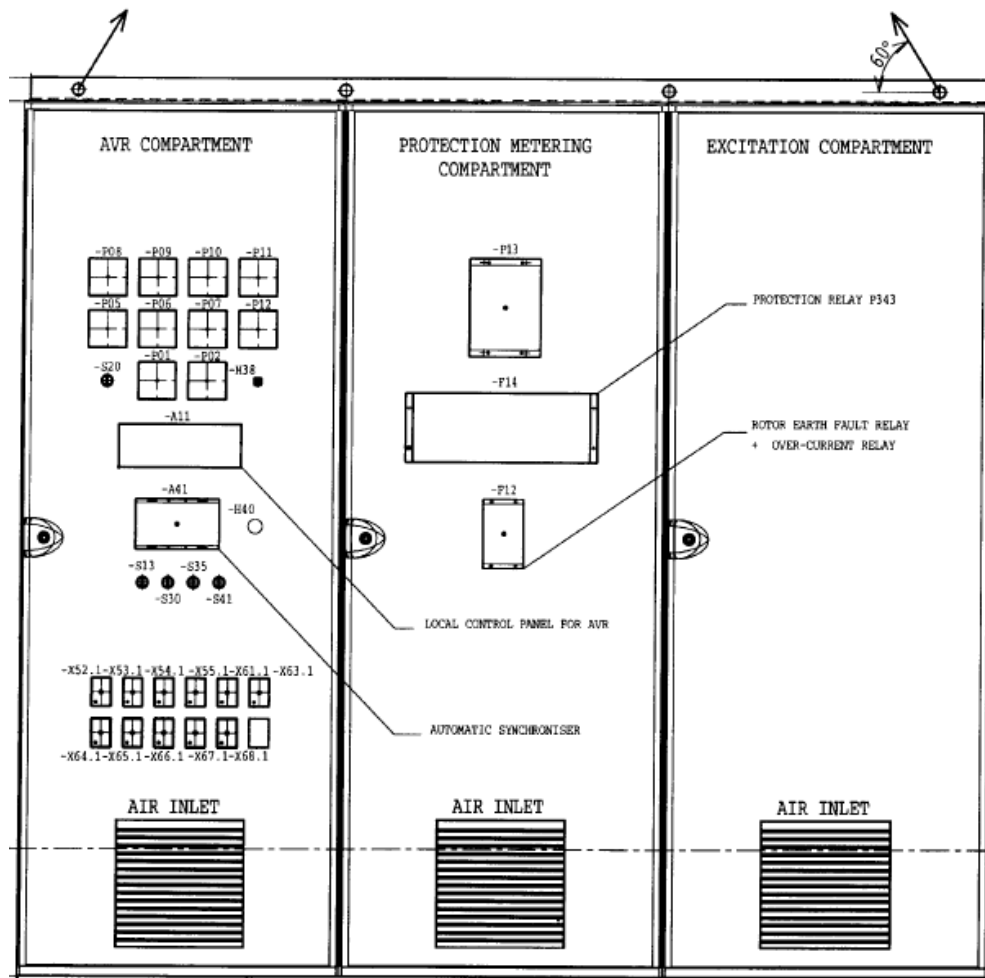
7A.7.1 Slinging principle with lifting eyes on the bottom

SLINGING PRINCIPLE





7A.7.2 Slings principle with lifting eyes on the top



7A.8 Installing the cubicle

Typical installation

The cubicle must be installed in a well aerated building to avoid overheating. Heat dissipated by the cubicle must be extracted from the building.

For a good cooling air flow, the cooling air outlets should be kept unobstructed and clean.

Leave enough space around the cubicle for safe service access.

Free areas around the cubicle

Door access : 1000 mm minimum

Panel access : 500 mm minimum

Preparing before installation

When the cubicle is installed at its final place, remove the plastic cover with a cutting tool. Be careful not to damage the cubicle's protective coat of paint. Remove the dessicant bags from the plastic cover. The dessicant (clay) can be discharged without any problem to the environment.

Alignment

The cubicle must stand vertically and plumb. The maximum permissible misalignment of the cubicle base is 10 mm on the longitudinal side. Doors and cover plates must not be jammed or warped.

Before the cubicle is definitely secured, all doors must be tested for correct opening and closing. It must be possible to open/close the doors without them being deformed.

Securing on the floor

The cubicle should be secured on the floor with appropriate devices depending on the floor type. The points for attachment on the foundations are defined on the layout drawing of the cubicle. Refer to the layout drawing to make sure that the foundation slab has been designed to support the mass of the cubicle.

7A.9 Wiring

7A.9.1 Necessary documents

Excitation cubicle layout drawing

Wiring diagram

(Refer to the list of drawings for the accurate number of documents).

7A.9.2 Safety measures

Before the first intervention on the cubicle, check that it has been correctly secured to the floor.

Operators should wear personal safety equipment in compliance with relevant local regulations, at least safety shoes and a helmet.

Personnel should be made aware of the danger of the lifting eyes when in the vicinity of the cubicle.

Work on the excitation cubicle should only be carried out by skilled persons.

Also check that no cables are live.

7A.9.3 Electrical connections

For electrical connections a bolted assembly system must be used. It is also advisable to fit a contact spring washer to NF F 25 511.

Tightening torques of the hexagon head bolts (class 8.8) :

Bolt size	M5	M6	M8	M10	M12	M16	M20	M24
Torque (Nm)	5.0	8.4	20	41	71	175	355	620

7A.9.4 Preparation before wiring

Before the equipment is wired, a visual inspection should be made. Check for correct securing of all internal devices and for absence of damage resulting from transportation and handling of the cubicle.

Remove the protective panels from all cable inlet and outlet openings. These panels are normally located at the bottom of the cubicle as shown below. In some cases, the cable inlet and outlet openings are at the top of the cubicle. Refer to the specific layout drawing for the locations of cable inlets and outlets. Unscrew the hexagon head bolts. Remove the protective panels and drill holes to fit cable glands (not supplied), consistent with the size of the cables.

7A.9.5 Wiring

Note: For wiring details, refer to the wiring diagram.

The control cables and the power cables must be correctly fitted and clamped on the fixing bar to avoid stressing the terminals.

Cable connections

All the cables enter the cubicle through an opening in the compartment at the bottom or the top of the cubicle. Drill holes into the protective panel to fit cable glands (not supplied) .

Take care to connect the shield of all analogue cables (twisted pair & RTD) to the grounding system

Don't mix power ,control reference and measure circuits in the same cable tray without sufficient spacing

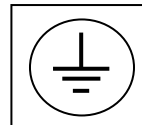
PE wiring

The main Protective Earth wire (PE) must be connected to a cooper bar in the bottom of the cubicle or to a welded stud outside the cubicle. Studs for earth connections are marked with letters "**PE**" and with an earth symbol.

Only yellow/green wires may be used for Protective Earth connections. However, It is not allowed to use the yellow/green wires for other connections.



Protective earth
Customer connection



Ground symbol

PE wire sizes

Size of Power supply wires S (mm ²)	Size of Protective Earth wires S _{PE} (mm ²)
$S \leq 16$	$S_{PE} = S$
$16 \leq S \leq 35$	$S_{PE} = 16$
$S > 35$	$S_{PE} = S/2$

The earth wire from the low voltage devices are connected to dedicated terminalof their terminal boards. Frames are connected to the structure of the cubicle.

The screening braids of a large components shielded cables must be connected to the cooper bar along the terminal boards.

Check the correct earthing of all doors and removable panels.

7A.10 Cleaning

Once the cubicle has been installed, clean it inside before power is applied.

Remove all tools, metallic parts, etc., from the cubicle. You can clean the inside of the cubicle with compressed air (be careful not to cause damage by using excessive pressure), or with a vacuum cleaner.